Assignment 4 - Normalization

Notation: <u>Bold underlined</u> = primary key; *italic* = foreign key; ***<u>bold, underlined, italic</u>*** = both PK and FK.

Exercise 1 - Happy Supplies Parts Order

1. Assumptions

- customerNo uniquely identifies a customer and determines customerName and customerType.
- partNo uniquely identifies a part and determines partName, partType, cageCode, and the unitPrice shown on the order.
- Because the source has no order number, the combination (customerNo, orderDate, orderTime) identifies one order. The same customer cannot place two orders at exactly the same date and time.
- A part occurs at most once on an order; quantityOrdered is the total ordered quantity for that part.
- employeeName is sufficient to identify the employee for this exercise. No employee identifier may be invented, and employees may serve any customer.
- Each part is stored in one cage and has one type and one current unit price.

2. Source form (UNF)

The handwritten form contains an order-level group and a repeating group of part lines. The repeating group prevents 1NF.

UNF relation	Attributes
partsOrder	customerNo, customerName, customerType, orderDate, orderTime, employeeName, {partNo, partName, partType, cageCode, quantityOrdered, unitPrice}

customerNo	customerName	customerType	orderDate	orderTime	employeeName	repeating part lines
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D. Harrison	(10654, Float Control, Plumbing, G413, 4, 12); (10456, Modulator, Electrical, H433, 3, 7); (10776, Hose Assembly, Plumbing, G413, 7, 9); (10657, Float Assembly, Plumbing, G413, 5, 10)

Exercise 1 - Normalization Steps

3. First Normal Form (1NF)

Remove the repeating group by placing one part line in each row. All values are now atomic. The composite PK is (customerNo, orderDate, orderTime, partNo).

customerNo	customerName	customerType	orderDate	orderTime	employeeName	partNo	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/24	10:30	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/24	10:30	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/24	10:30	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/24	10:30	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

Functional dependencies

- customerNo -> customerName, customerType
- partNo -> partName, partType, cageCode, unitPrice
- (customerNo, orderDate, orderTime) -> employeeName
- (customerNo, orderDate, orderTime, partNo) -> quantityOrdered

4. Second Normal Form (2NF)

The 1NF key is composite, and several non-key attributes depend on only part of it. Remove those partial dependencies into customer, part, and partsOrder. quantityOrdered remains in partsOrderLine because it depends on the entire original key.

2NF relations	Attributes
customer	**customerNo**, customerName, customerType
part	**partNo**, partName, partType, cageCode, unitPrice
partsOrder	***customerNo***, **orderDate**, **orderTime**, employeeName
partsOrderLine	***customerNo***, ***orderDate***, ***orderTime***, ***partNo***, quantityOrdered

Exercise 1 - Final 3NF Design

5. Third Normal Form (3NF)

No remaining non-key attribute determines another non-key attribute. Therefore the 2NF relations are already in 3NF. The order line references its order through a three-column composite FK.

3NF relations	Attributes
customer	**customerNo**, customerName, customerType
part	**partNo**, partName, partType, cageCode, unitPrice
partsOrder	***customerNo***, **orderDate**, **orderTime**, employeeName
partsOrderLine	***customerNo***, ***orderDate***, ***orderTime***, ***partNo***, quantityOrdered

Foreign-key references

- partsOrder.customerNo -> customer.customerNo
- partsOrderLine.(customerNo, orderDate, orderTime) -> partsOrder.(customerNo, orderDate, orderTime)
- partsOrderLine.partNo -> part.partNo

> **Why the design is 3NF:** every non-key attribute is dependent on the key, the whole key, and nothing but the key. Customer facts are stored once, part facts are stored once, order facts are stored once, and line quantity is stored only at the order-part intersection.

Exercise 2 - Panacea Mental Health

1. Assumptions

- staffNo uniquely identifies a therapist and determines therapistName.
- patNo uniquely identifies a patient and determines patName.
- A therapist cannot conduct two appointments at the same date and time, so (staffNo, appointmentDate, appointmentTime) uniquely identifies an appointment.
- The narrative establishes (staffNo, appointmentDate) -> branchNo: on a given date, a therapist sees patients at exactly one branch.
- A patient may have multiple appointments in one day, including with different therapists; therefore patNo and date alone are not a key.
- Branch details beyond branchNo are outside the supplied data and are not invented.

2. Source table and 1NF

The source is already in 1NF because each cell is atomic and there are no repeating columns. Its PK is (staffNo, appointmentDate, appointmentTime).

staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Functional dependencies

- staffNo -> therapistName
- patNo -> patName
- (staffNo, appointmentDate) -> branchNo
- (staffNo, appointmentDate, appointmentTime) -> patNo

Exercise 2 - From 1NF to 3NF

3. Second Normal Form (2NF)

Remove partial dependencies on staffNo and on (staffNo, appointmentDate). patName remains with appointment at this stage because it is transitively dependent through patNo, not partially dependent on the composite PK.

2NF relations	Attributes
therapist	**staffNo**, therapistName
therapistWorkDay	***staffNo***, **appointmentDate**, branchNo
appointment	***staffNo***, ***appointmentDate***, **appointmentTime**, patNo, patName

4. Third Normal Form (3NF)

In appointment, patNo -> patName is a transitive dependency from the appointment key. Move patient facts into patient. All remaining determinants are candidate keys of their relations.

3NF relations	Attributes
therapist	**staffNo**, therapistName
patient	**patNo**, patName
therapistWorkDay	***staffNo***, **appointmentDate**, branchNo
appointment	***staffNo***, ***appointmentDate***, **appointmentTime**, *patNo*

Foreign-key references

- therapistWorkDay.staffNo -> therapist.staffNo
- appointment.(staffNo, appointmentDate) -> therapistWorkDay.(staffNo, appointmentDate)
- appointment.patNo -> patient.patNo

> **Result:** therapist names and patient names are each stored once, a therapist's branch assignment is stored once per day, and each appointment records only the therapist/date/time key plus its patient.

Exercise 3 - Maid Better Contract Hours

1. Assumptions

- eNo uniquely identifies an employee and determines eName.
- contractNo uniquely identifies a contract. Each contract applies to exactly one event, so contractNo determines eventNo.
- eventNo uniquely identifies an event and determines eventLoc.
- An employee has at most one hours total per contract; therefore (eNo, contractNo) is the source table PK.
- hours represents the employee total for that contract rather than separate daily time entries.
- Different contracts may reference the same event, as demonstrated by C1024 and C1026.

2. Source table and 1NF

The source table is already in 1NF: all values are atomic. The composite PK is (eNo, contractNo).

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Functional dependencies

- eNo -> eName
- contractNo -> eventNo
- eventNo -> eventLoc
- (eNo, contractNo) -> hours

Exercise 3 - From 1NF to 3NF

3. Second Normal Form (2NF)

eName depends on only eNo, and eventNo/eventLoc depend on only contractNo. Move these partial dependencies out of the composite-key relation. eventLoc may remain in contract at 2NF because the dependency through eventNo is transitive, not partial.

2NF relations	Attributes
employee	**eNo**, eName
contract	**contractNo**, eventNo, eventLoc
contractHours	***eNo***, ***contractNo***, hours

4. Third Normal Form (3NF)

In contract, contractNo -> eventNo -> eventLoc is a transitive dependency. Move eventLoc into event and retain eventNo in contract as a foreign key.

3NF relations	Attributes
employee	**eNo**, eName
event	**eventNo**, eventLoc
contract	**contractNo**, *eventNo*
contractHours	***eNo***, ***contractNo***, hours

Foreign-key references

- contract.eventNo -> event.eventNo
- contractHours.eNo -> employee.eNo
- contractHours.contractNo -> contract.contractNo

> **Result:** employee facts, event facts, contract-to-event facts, and employee hours per contract are each represented exactly once at the appropriate level of detail.

Final validation

- All three exercises reach 3NF.
- Every relation is named in singular lower camel case.
- No surrogate keys or new attributes were introduced.
- PKs and FKs use the required visual formatting.